



Technische
Universität
Braunschweig

**Decision
Support**

Institut für Wirtschaftsinformatik



Data Driven Decision Making

Lecture 10 – Combined Methods

Agenda

- Recap Policy Classes
- Combinations
- Case Study: Dynamic Relocations in Bike-Sharing Systems
- Achievements

Policy Classes (Compare Powell 2011)

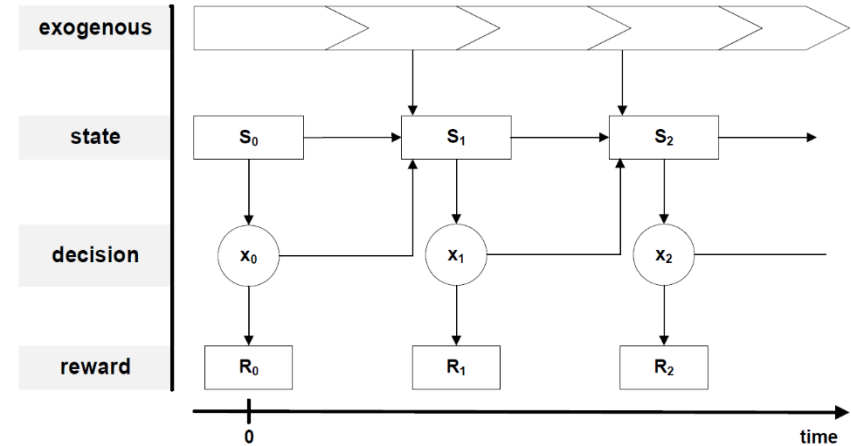
- Policy Function Approximation
- Rolling Horizon Re-Optimization
- Cost Function Approximation
- Look-ahead Models
- Value Function Approximation

Rule-Based

Optimization-Based

Policy Classes I

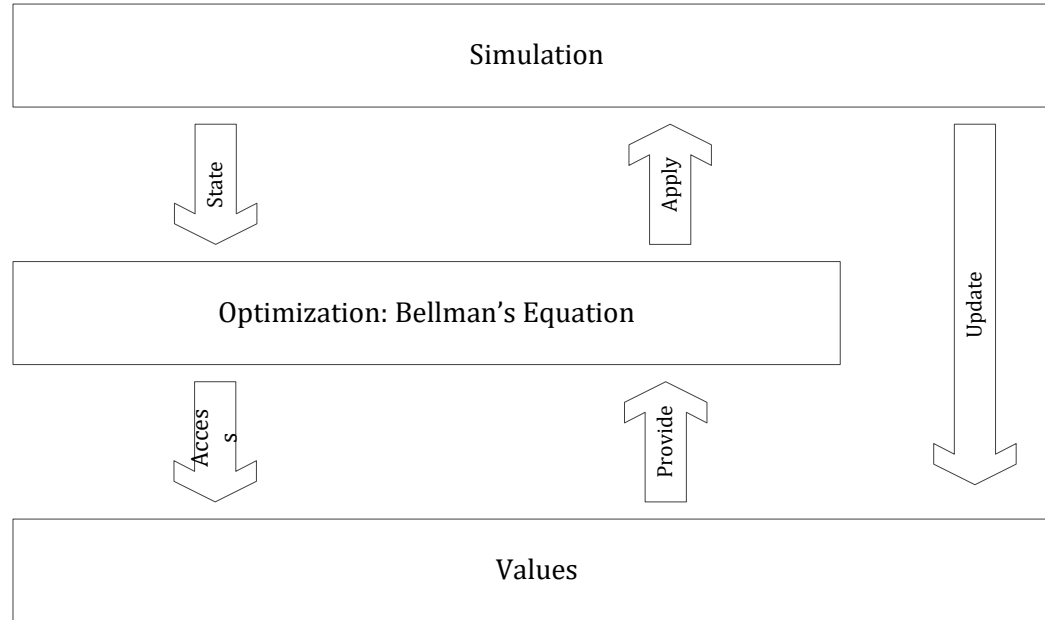
- Policy function approximation:
Transfer a rule-of-thumb into a policy
- Rolling-Horizon Re-Optimization:
Determine decision based on current information (myopic)
- Cost-Function Approximation:
Manipulate reward function or constraints to incentivize (/avoid) (in)flexible decisions
- None of the classes anticipates the future (information & decision model) explicitly



Policy Classes II: Lookahead

- Methods consider future changes via sampling of realizations
- Rollout algorithm:
 - Simulate paths of sampled realizations and decisions made by a base policy
 - Use simulated values with Bellman Equation
- Multiple Scenario Approach:
 - Sample sets of realizations (scenarios)
 - Solve scenarios individually
 - Determine most similar solution → decision
- Disadvantages: runtime, decision making in simulation

Value Function Approximation (VFA)



Soeffker et al., 2016

Disadvantages of ADP-Methods

- Rolling horizon does not anticipate future developments
- Policy function approximations do not exploit full detail of state and stochastic information
- Cost function approximations are difficult to tune and ignore specifics in state and stochastic information
- Lookahead methods usually require significant runtime and can only capture a few samples and decisions
- Value function approximations require aggregation and storage of results, scalability issues
- Further issues such as intuition, interpretability, etc.

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- **Combinations**
- Case Study: Dynamic Relocations in Bike-Sharing Systems
- Achievements

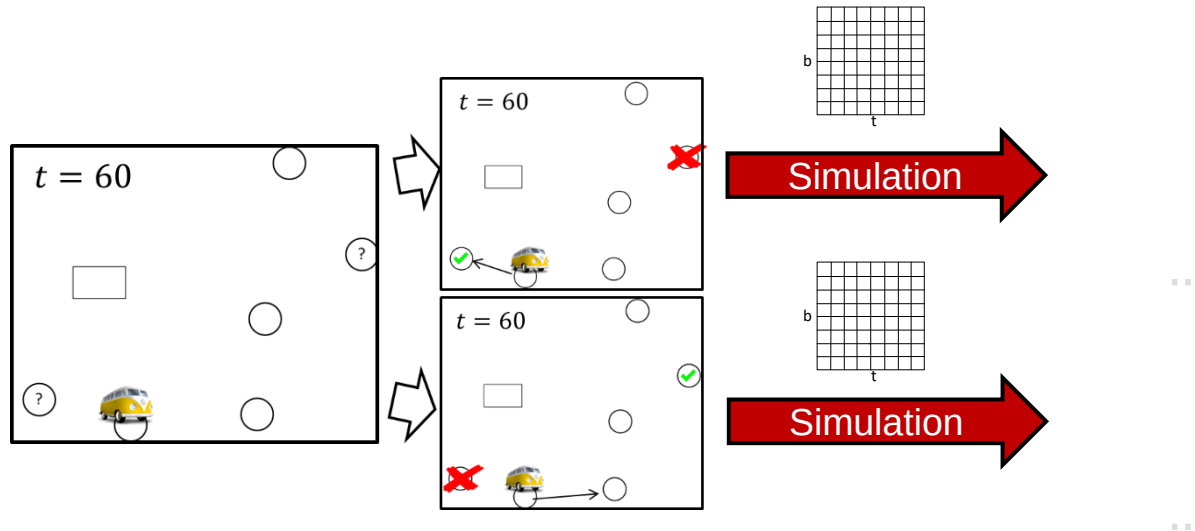
We can combine ADP-methods

- Combinations of ADP-methods can exploit their strengths and alleviate their shortcomings
- For example, integrate PFA, CFA, or VFA into a rollout algorithm
 - PFA or CFA can replace myopic base policy in rollout algorithm
 - Even VFA can evolve an improved base policy for rollout
 - The simulation horizon may be split into an 1. rollout and 2. VFA part

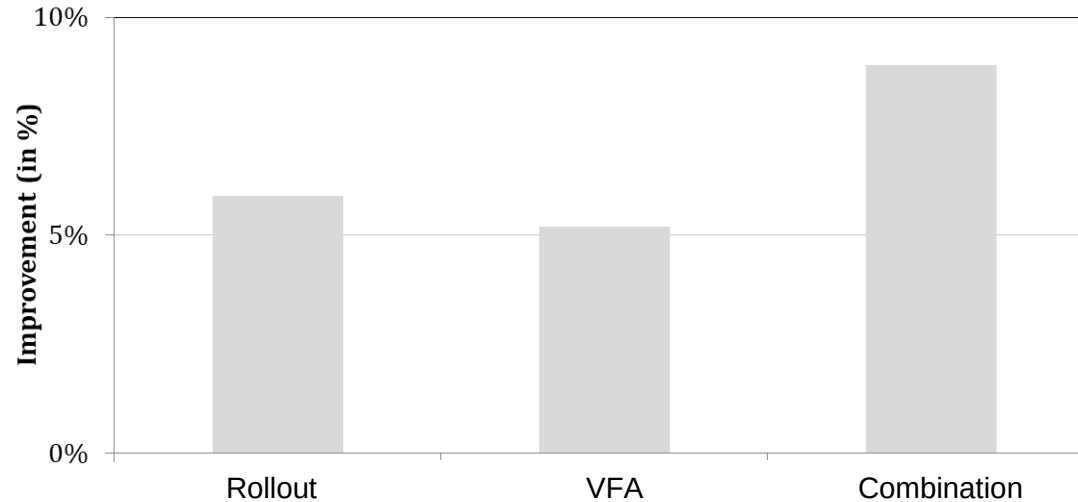
Example: Dynamic Routing

$$X_k^{\pi^*}(S_k) = \arg \max_{x \in X(S_k)} \{R_k(S_k, x) + V(S_k^x)\}$$

- Detailed online simulations with rollout
- Anticipatory base policy induced by offline VFA-values



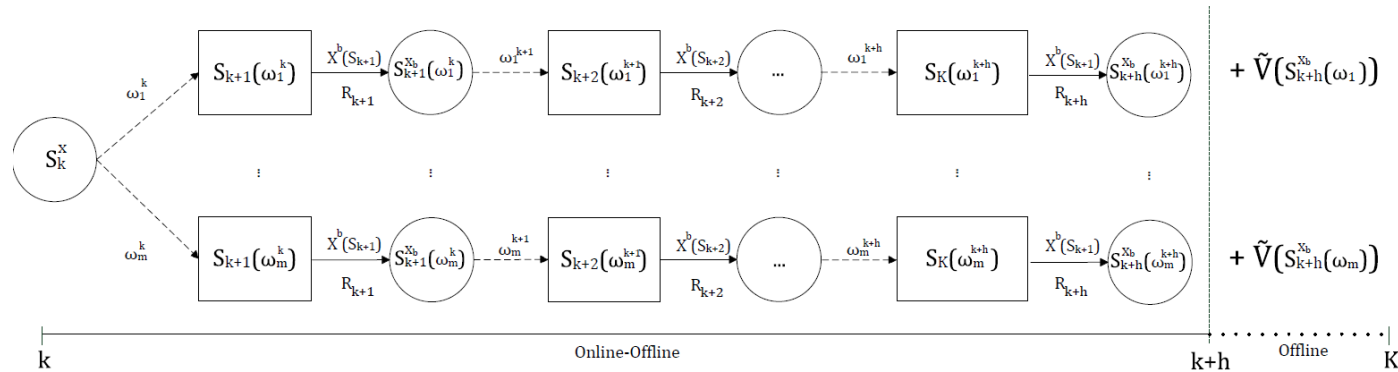
Results (Benchmark: Anticipatory Insertion (Ghiani et al. 2012))



Ulmer, Goodson, Mattfeld, Hennig: Offline-Online ADP for Dynamic Vehicle Routing with Stochastic Requests, Transportation Science, 2019

Limited Lookahead Horizons + VFA

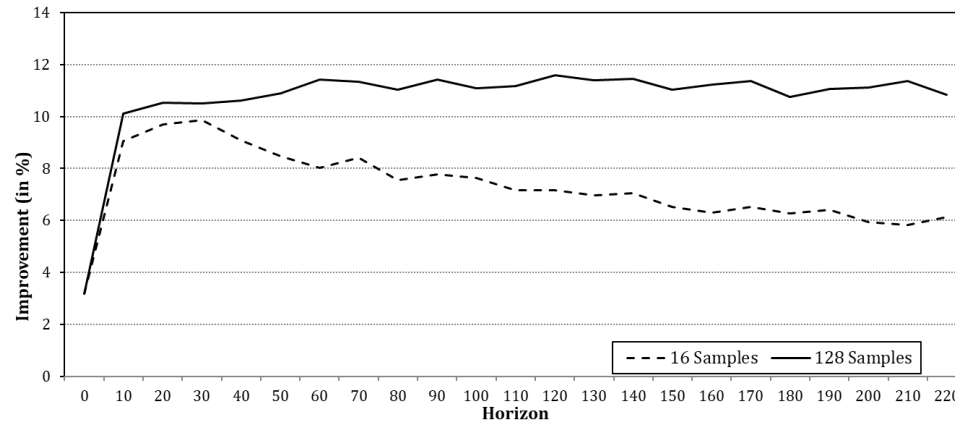
- Apply rollout for h decision points
- Add VFA-value of last simulated post-decision state
- Avoid discrepancy of rollout and reality, save runtime



- Ulmer, Horizontal combinations of online and offline approximate dynamic programming for stochastic dynamic vehicle routing, Central European Journal of Operations Research, 2019

Example: Dynamic Vehicle Routing

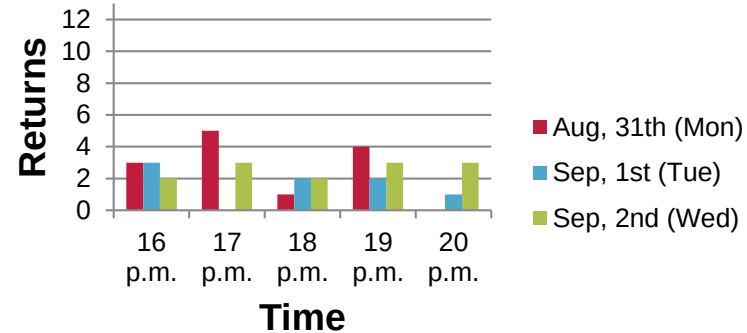
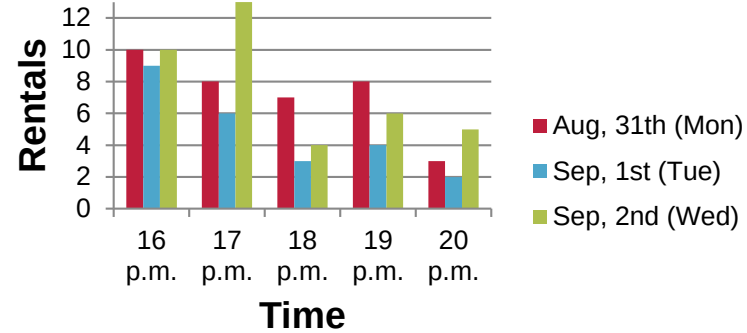
- Comparison against myopic policy
- Improvement for rollout algorithm with 16 and 128 samples and different horizons h
- With 16 samples, limiting horizon to 30 points provides best solutions



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- **Case Study: Dynamic Relocations in Bike-Sharing Systems**
- Achievements

Ad-hoc control of bike relocation

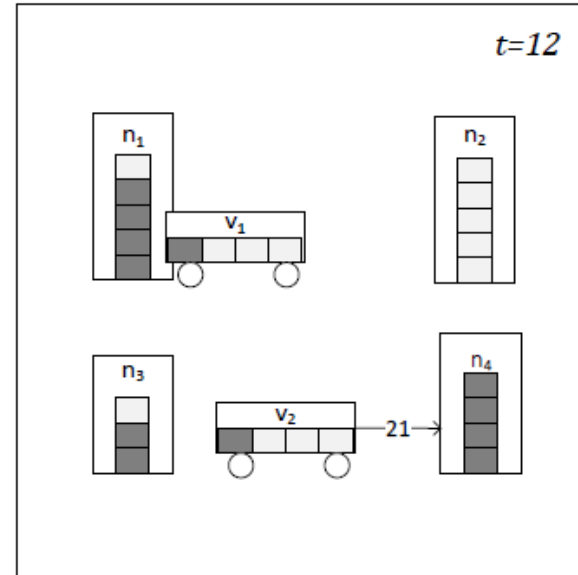


Ad-hoc control of bike relocation

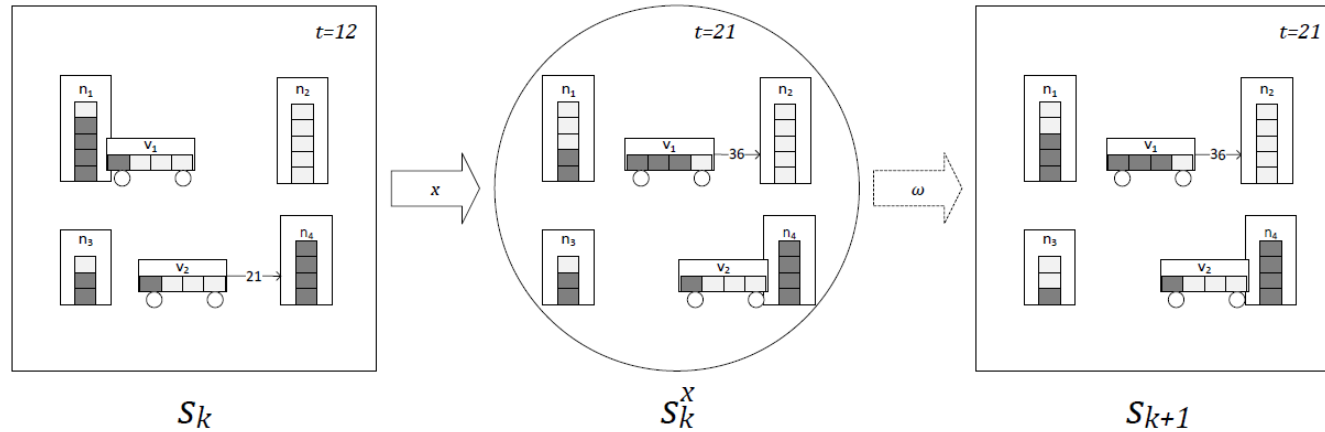
- Control of the decisions of a bike relocation truck
- Alternating decisions
 - At a station, decide on the number of bike to (un-)load
 - Once done (un-)loading, identify station with urgent need to go
- Urgent need is defined by missing bikes or missing bike racks
- Missing rack or missing bike define so called service fail
- One aims at the overall minimization of service fails over time
- In order to take decisions, a measure for the expected future fail of service is needed

Problem Definition

- Station-based bike sharing system
- Time horizon
- Stations with capacity and initial fill levels
- Fleet of vehicles (1 in the simplest case)
- Stochastic customer trips (known probability)
- Dynamic routing and loading
- Objective: Minimize expected failed demands (pick up and drop off)



Markov Decision Process

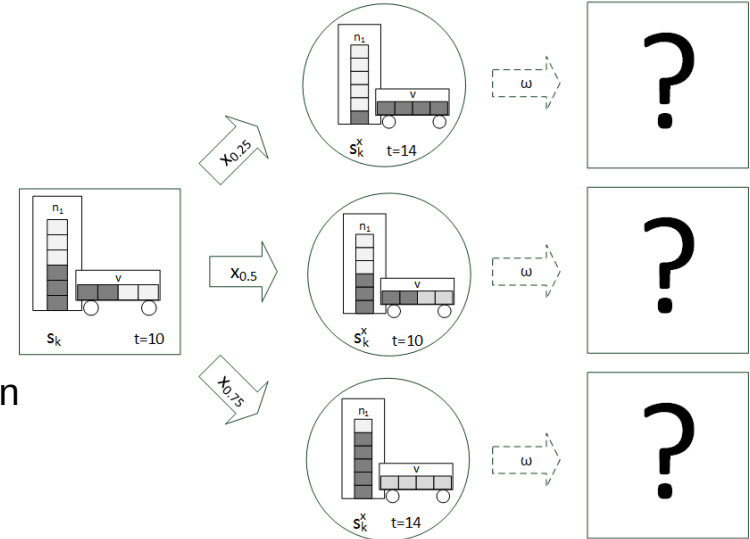


Markov Decision Process

- Decision Point: Vehicle arrives at station
- States:
 - Vehicles (next stations and arrival times)
 - Stations (current fill levels)
- Decisions:
 - Loading at current station
 - Next station to visit
- Post-decision state: as state but with updated vehicle info and station fill level
- Stochastic information: New demand for return and pickup
- Transition:
 - New state when next vehicle arrives
 - Updated fill level based on demand
 - Observed failed demand

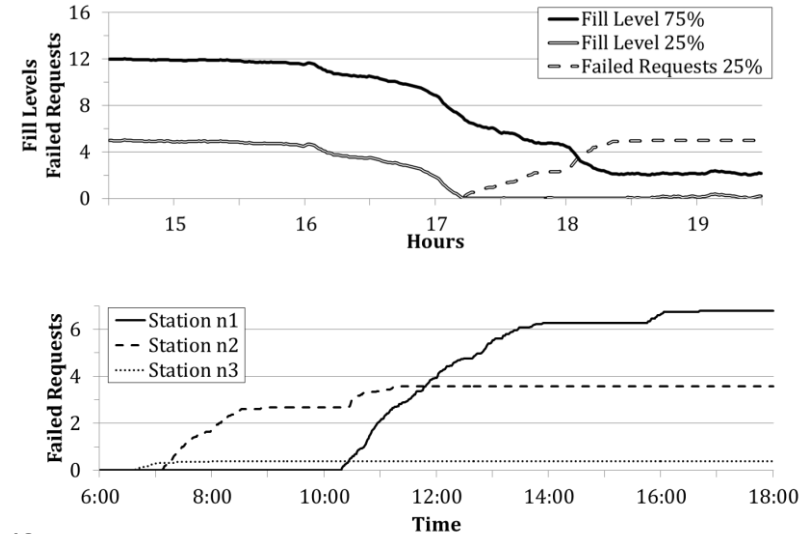
Dynamic Lookahead Policy

- Loading:
 - Generate candidate solutions for current station (via PFA, 25%, 50%, 75%)
 - Simulate future trips (customer demand) to evaluate candidate solutions
 - Select fill level with max prevented failed demand
- Routing:
 - Use same simulation to determine next station based on
 - Max prevented failed demand
 - Potential of other vehicles of the fleet (assignment problem)



Determine expected failed demand

- Consider to fill a station to 25% or 75%
- Simulation of future demand leads to failed requests for the 25% fill level
- Fill levels do not differ for the next 2½ hours
- Compare development of failed requests for three independent stations (n1, n2, n3)
- n3 receives one failed request early
- n2 receives increasing # of failed requests
- n3 receives many failed requests late in time
- On the long run the failed requests do not differ anymore
- Use VFA to learn hour-of-the-day dependent simulation horizons

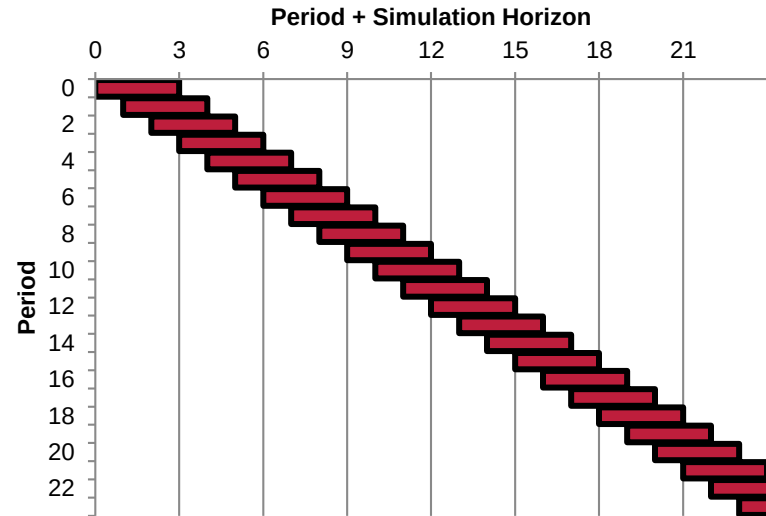


Dynamic Lookahead Horizon

- Subdivide horizon T into periods
- Consider lookahead horizons Δ_p
- A policy can be represented by a 24-dimensional vector, e.g. (3,3,3...,3,2,1)
- How to determine an appropriate policy?
- Short horizon neglect fails in the far future
- Long horizon may does not relate to current decision anymore $\rightarrow$ white noise

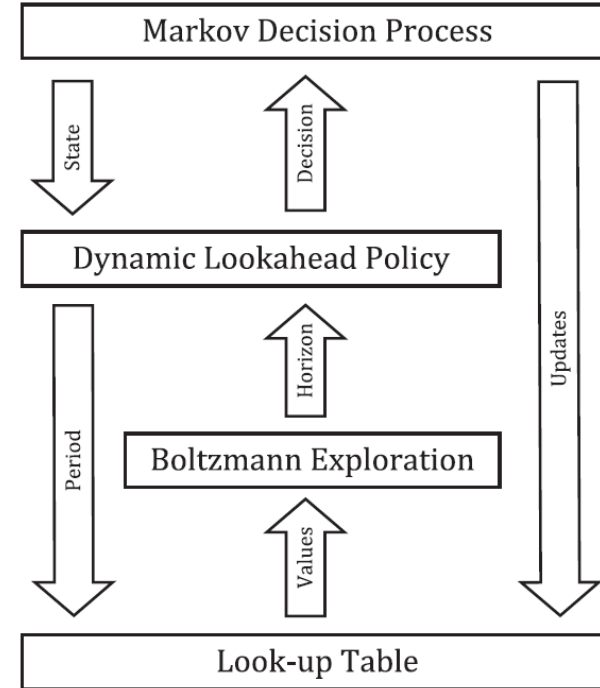
$$\{0, \dots, 23\} = P$$

$$\{0, 1, \dots, 6\} = \Delta$$



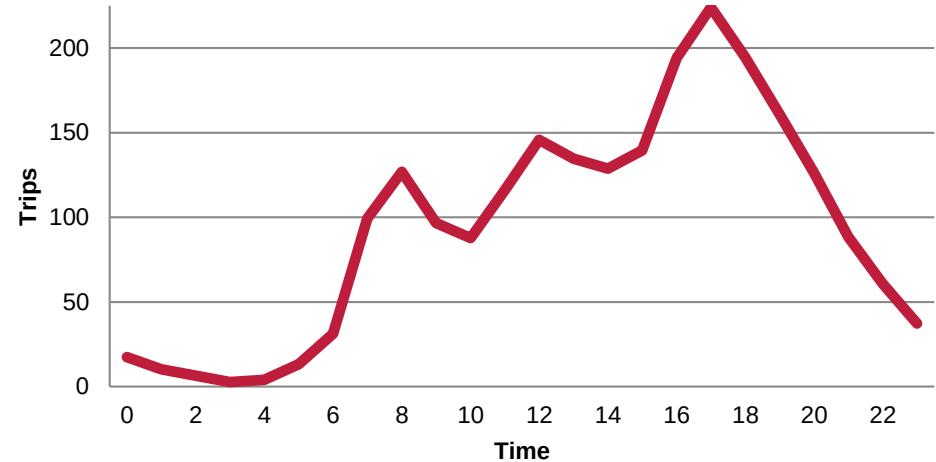
Value Function Approximation

- Lookup-table with values
- 24 (hours) x 7 (horizons) entries
- Value → the expected failed demand if applying a horizon in certain hour
- Start with initial values
- Simulate MDP
- In state, determine horizon from lookup-table (+ Boltzmann exploration)
- Apply lookahead with horizon
- After simulation run, update values in table



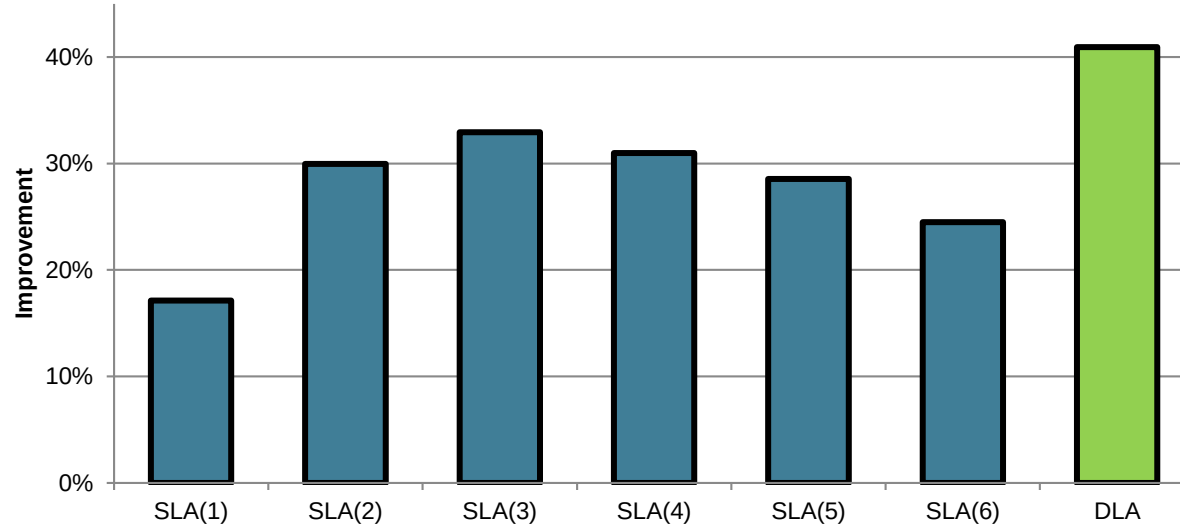
Case Studies: Instances

- Minneapolis' BSS „Nice Ride MN“
 - #trips show peaks at 8am, 12am, 5pm
 - Failed trips will occur during peak hours
 - Guess: incorporate next peak in horizon
-
- Benchmark algorithms:
 - PFA (safety buffer approach)
 - Static fixed horizon of n hours: SLA(n)

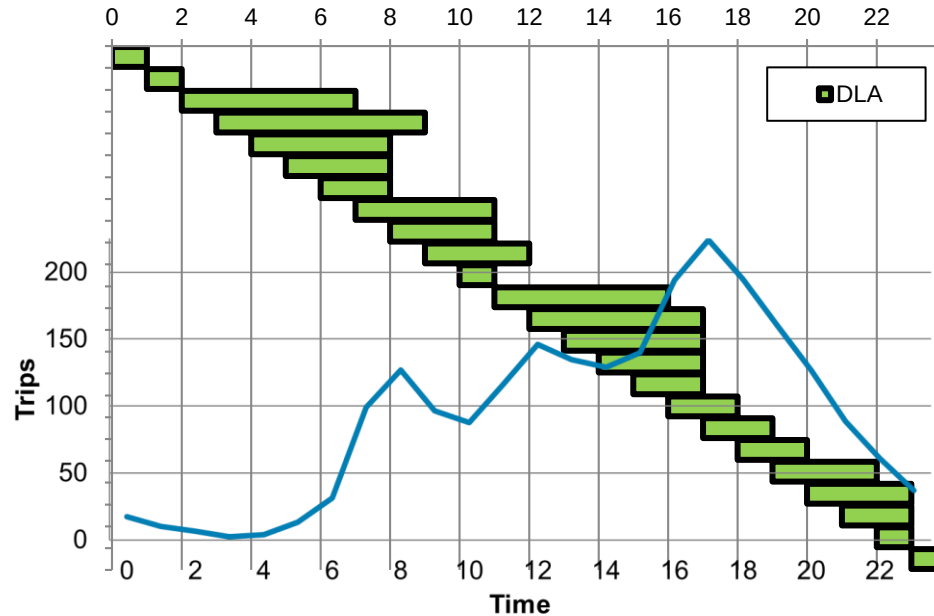


Case Studies: Results

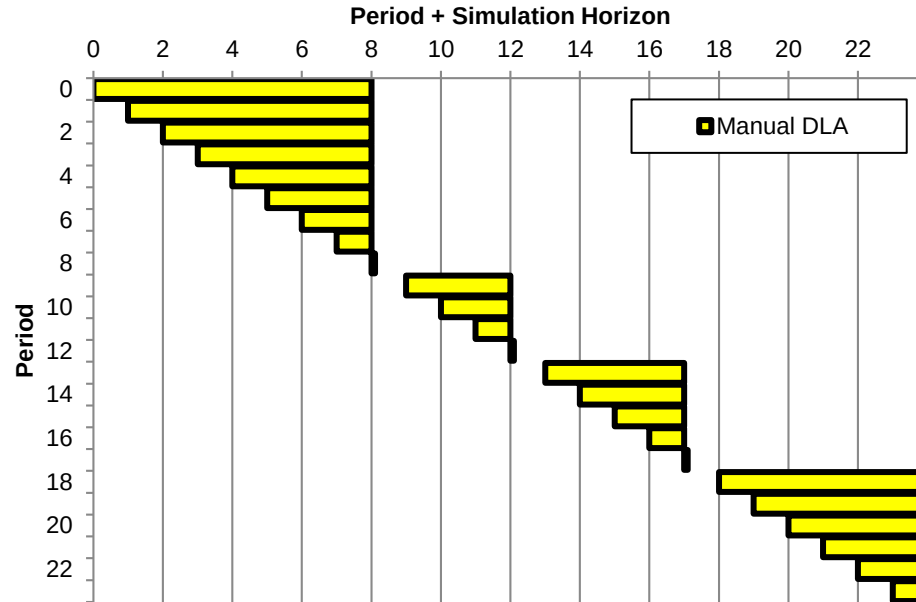
Reduction of failed demand against PFA



Case Studies: Dynamic Simulation Horizons

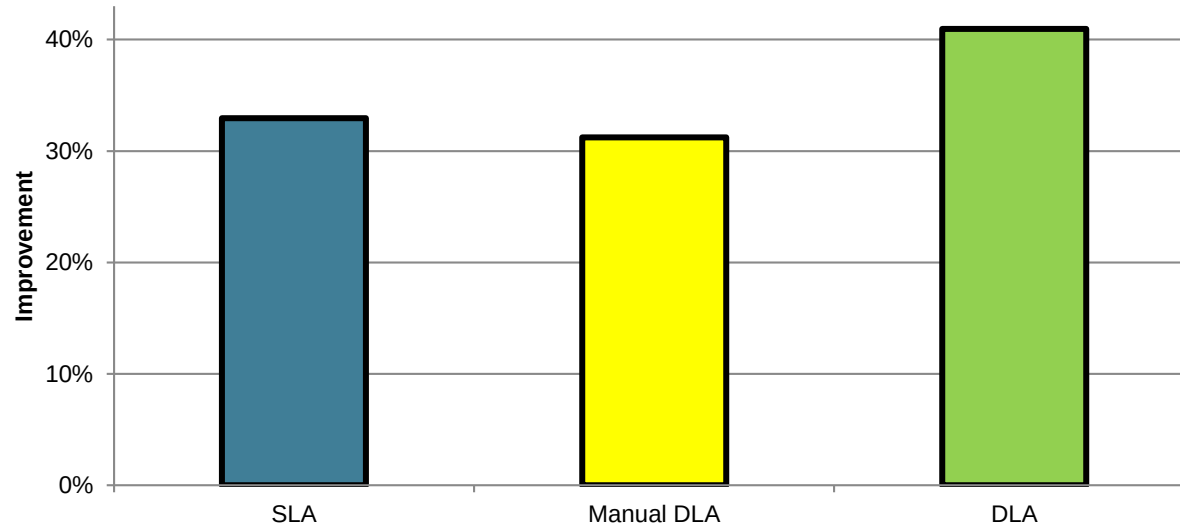


Case Studies: Dynamic Simulation Horizons



Case Studies: Results

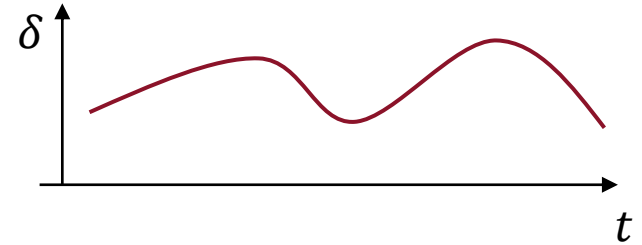
Reduction of failed demand against PFA



VFA-Simulation captures “hidden” problem and data characteristics

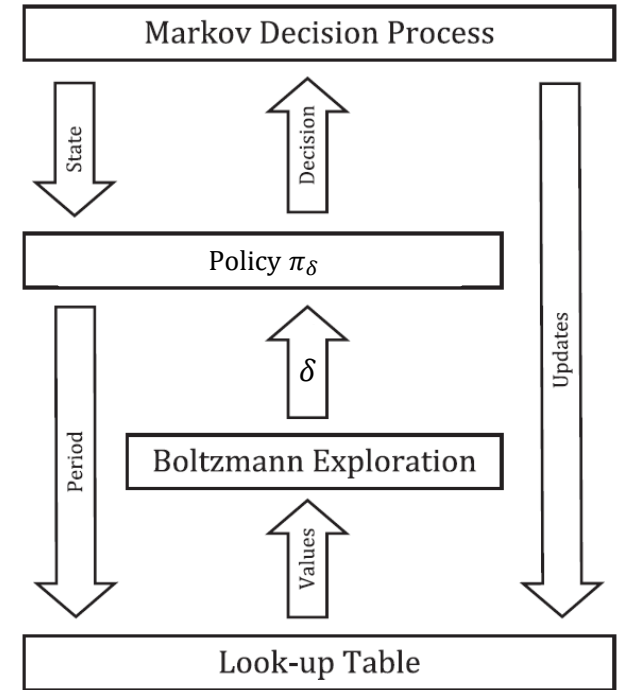
General Procedure

- Assume a policy π_δ with δ a tuning parameter
- For example, δ defines a threshold (safety buffer, maximal detour, lookahead horizon)
- We have a set of possible values for $\delta \in \Delta$
- We can define a policy class $\Pi^c = \{\pi_\delta^c: \delta \in \Delta\}$
- A different policy (δ) could be beneficial for different states
- Dependent on demand, resources, and environment
- Example: different δ for different hours of the day



General Procedure

- Nearly infinite number of combinations
- Divide time horizon into a set of periods
- Learn value $V(p, \delta)$ for every period p and parameter δ with VFA.
- Store values in lookup table.
- Instead of Bellman Equation, apply Boltzman Exploration to force exploration of solution space
- Our VFA is not used to select a decision but a policy



Generalization (still open)

- Learning state-dependent parametrizations of PFA, CFA, Lookaheads, etc.
- Parameters may vary not only over time, but also based on other observations:
 - Workload
 - Urgency
 - Etc. (compare features in lecture 9)

Summary and Outlook

- Anticipation is beneficial in controlling bike sharing systems
- Lookahead horizons should be “data-dependent”
- General method for large scale dynamic decision problems:
 - Parametrizable policy / Set of different policies
 - Learning state-dependent parameters (still open)
- In combination with PFAs, the procedure can maintain the intuition and interpretability of the method

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- **Achievements**

Data Driven Decision Making (agenda)

No.	Topic
1	Different states come with different data
2	Per state data can vary stochastically
3	Too much variation requires dynamism
4	Depiction by markov decision process
5	Optimal derivation of expected future value
6	Policy/cost function approximation depict value externally
7	Look-ahead methods cope with value internally
8	Value function approximation learns value
9	Data analytics is a key feature to VFA
10	Combination of methods yield further improvement

Data Driven Decision Making (theory)

- Data becomes important if
 - it changes over time and requires a temporal consideration
 - changes are stochastic or even distributions are unknown
 - variations are so large that information gain becomes important
- Dynamic decision making comes into play
 - an external stochastic process has to be modeled
 - an internal decision process has to be modeled
 - Both are tied by succeeding state representations
 - Markov process sequentially models stepwise
- Policy making by
 - Policy function approx., cost function approx., look ahead, value func. approx.
 - There is no favorite technique, application relates to curses of dimensionality

Data Driven Decision Making (application)

- A priori decision making has its advantages
 - Optimality may be warranted
 - Setup time for implementation
 - Communication of solution to stakeholders
- Dynamic decision making in the face of significant information gain
 - Assignment of internal resources to external demand
 - Current decisions impact future flexibility
 - Anticipation incorporates expected future reward
- Decision making combines
 - Optimization (first term of Bellman equation)
 - Analytics (second term of Bellman equation)
 - Combination of searching for unknown features and conclusion of statistical correlation

Outlook to next and final lecture

- The 11th lecture is devoted to practical food delivery
- It is based on white papers published by Uber eats
- The lecture focuses on data analytics and decision making
- Achievements of IDA, PMT and 3DM are summarized